

Fig. 2: An actual-size, single-side PCB for the solar-charged emergency light

The LED light is made using 30 white LEDs, while a 12V DC motor may be used for building a fan. Separate switches are used to switch on/off the light and the fan. Resistors R1 through R8, R16 and R17 are used to limit the current passing through white LEDs.

Battery level can be monitored through LED33 and LED34. LED33 glows when the battery voltage is less than the preset minimum value. LED34

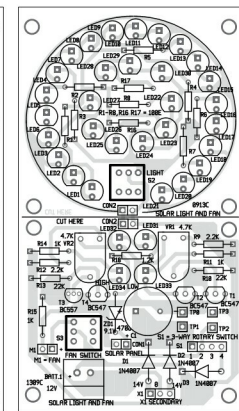


Fig. 3: Component layout for the PCB

glows to indicate that the battery has been charged up to the preset maximum value. Presets VR1 and VR2 are used to set the low-voltage and high-voltage levels of the battery.

Working of the circuit is simple. Keep switch S1 at position 3 to use the light and fan. The battery will discharge slowly. When the battery voltage goes below the preset minimum voltage level, LED33 will glow.

Disconnect both, the light and the fan, at this point to avoid deep discharging of the battery. To recharge the battery, check the status of LED31 and LED32. If these LEDs are glowing, it means the solar voltage is enough for charging. Turn switch S1 to position 2 and charge the battery using solar power. If the solar voltage is insufficient, turn the switch to position 1 and charge the battery directly through mains.

Construction and testing

An actual-size, single-side PCB for the solar emergency light is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable case. You can make separate assemblies for the LED light and the charger section, if you like, by cutting the PCB as indicated in its layout. Fix switches S1 through S3 on the rear side of panel, along with LED31 through LED34 that indicate various statuses. Connect the mains supply, battery and the solar panel to the PCB at suitable places meant for them.

To check the circuit for proper functioning, verify various voltage levels mentioned in the test point table. ●

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